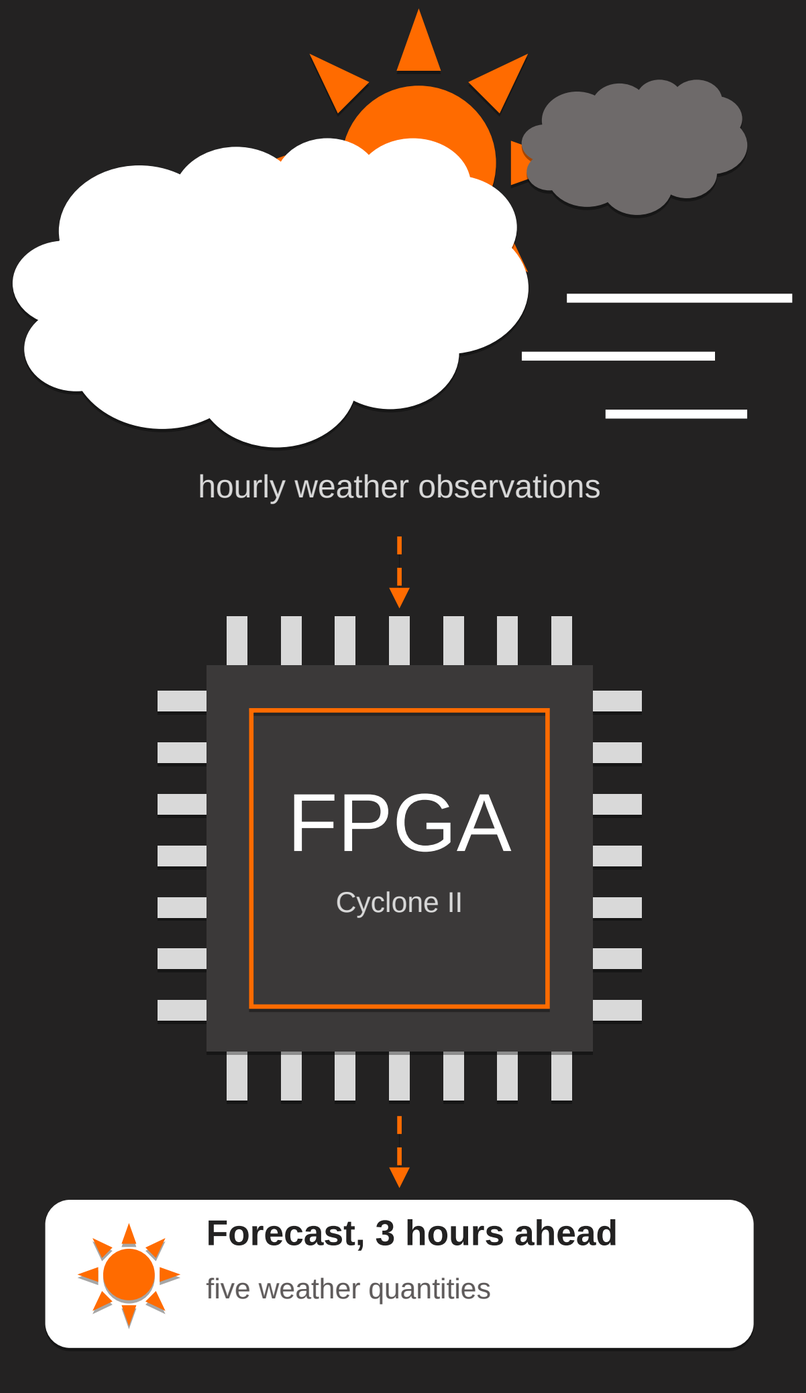
WEATHER FORECASTING

ON FPGA

Fixed-point inference of five weather quantities, three hours ahead, on a Terasic DE2 Cyclone II board

CE436 · DISCRETE-TIME SIGNAL PROCESSING

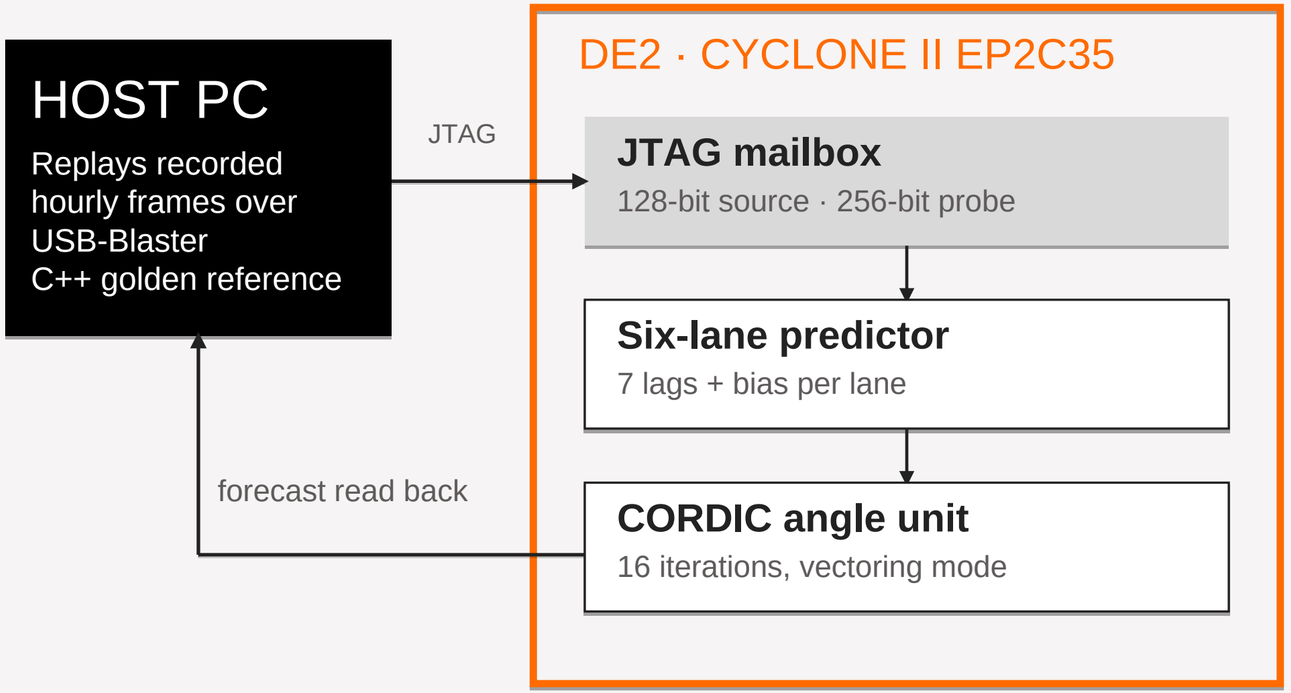
Final defense



WHAT WE FORECAST

Five quantities, three hours after the latest recorded hourly frame

01	Temperature	°C
02	Surface pressure	kPa
03	Relative humidity	%
04	Wind speed	m/s
05	Wind direction	degrees



Outside our scope: training on the FPGA, live sensors, rainfall probability, LED or LCD output.

The 3-hour horizon belongs to the data timestamps. Replaying frames faster than real time does not change it.

FROM ONE QUANTITY TO FIVE

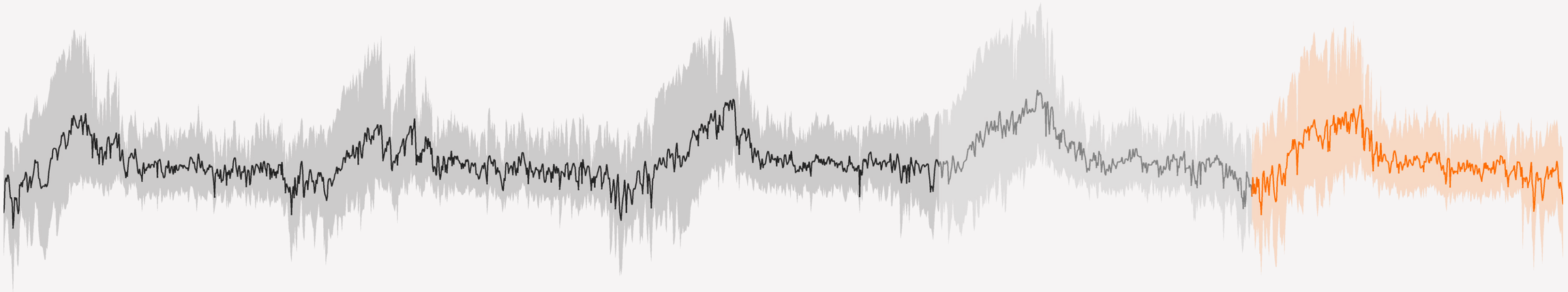
The temperature prototype set up the workflow; the weather design changed the model and the datapath

	Temperature prototype	Current weather design
Model	$\hat{T}(t+3) = T(t-21) + a\cdot\Delta\text{daily} + b\cdot\Delta\text{recent} + c$	$\hat{x}(t+3) = \text{bias} + \Sigma\, w_k \cdot x(t-\ell_k),\ \ell \in \{0,1,3,6,12,21,24\}$ h
Learned parameters	3 (a, b, c)	7 weights + 1 bias per lane, 6 lanes
Outputs	Temperature	Temperature, pressure, humidity, wind speed, direction
Arithmetic	One lane, signed 16-bit Q8	Six lanes, signed 18-bit Q8 samples, Q14 weights
Wind direction	Not modelled	Cosine and sine lanes + 16-iteration CORDIC
Test MAE, temperature	0.538 °C	**0.508 °C**

Unchanged: hourly input frames, the three-hour horizon, offline training, and the C++ bit-exact reference flow.

CHRONOLOGICAL EVALUATION

Fit on 2021–2023, choose on 2024, report once on 2025



Daily range and daily mean of the hourly 2 m temperature, Ho Chi Minh City (NASA POWER, UTC)

2021–2023

26,253

Training examples

Fits the coefficients

2024

8,781

Validation examples

Chooses lags and ridge penalty

2025

8,757

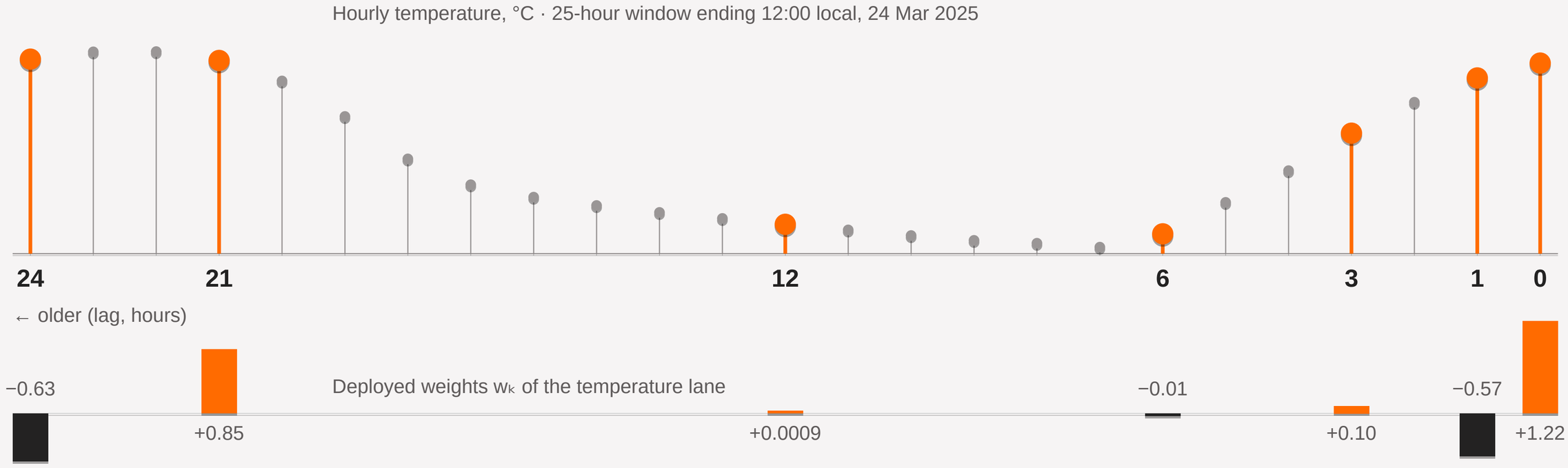
Test examples

Reported once, model frozen

SEVEN-LAG REGRESSION

Each lane predicts its own variable from seven past values and a bias

$$\hat{x}(t+3) = \text{bias} + \Sigma_k \; w_k \cdot x(t - \ell_k) \qquad \ell \in \{0, 1, 3, 6, 12, 21, 24\} \text{ h}$$



Temperature lane: deployed float weights over a real 2025 test window. The largest weights sit at 0–1 h and 21–24 h.

FIXED-POINT DATAPATH

Keep full precision through the sum; shift once at the end

Sample
`18-bit, Q8`

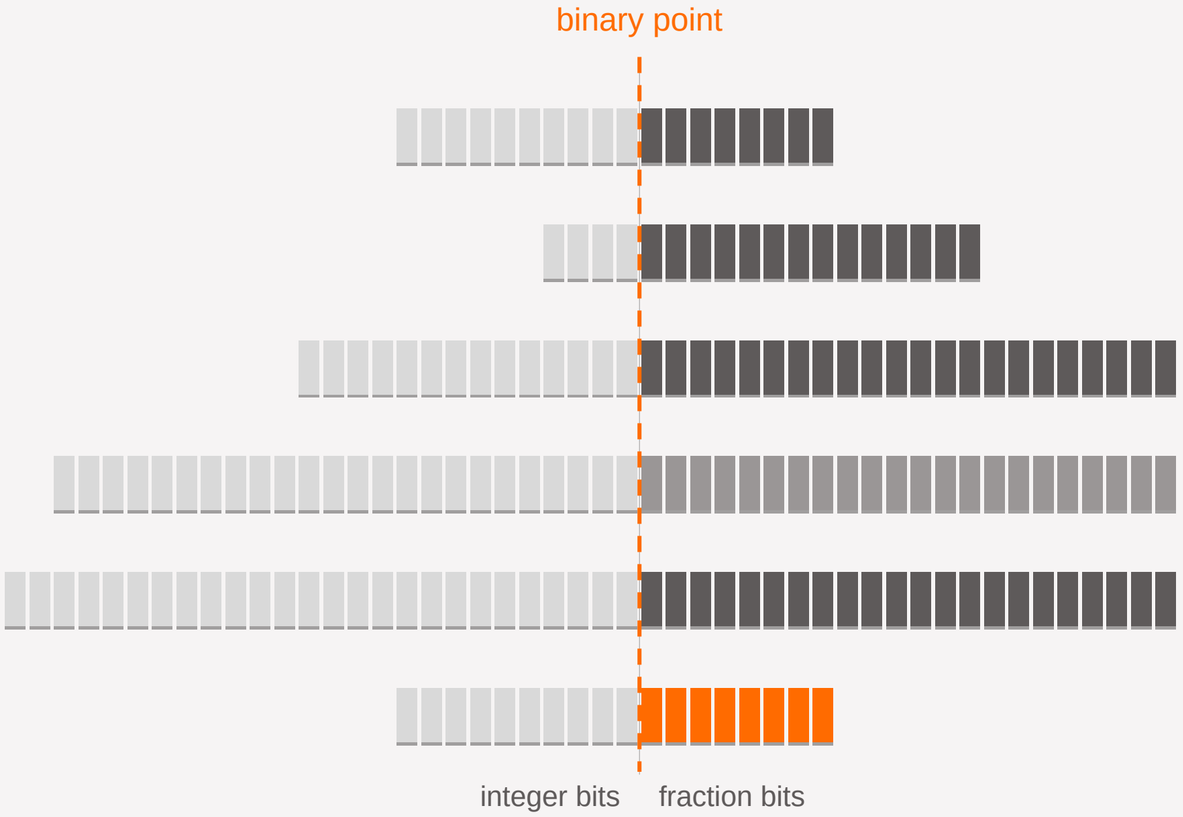
Weight
`18-bit, Q14`

Product
`36-bit, Q22`

Bias, aligned
`32-bit Q8, × 2^14`

Accumulator
`48-bit, Q22`

Output
`18-bit Q8 after >>> 14`



AFTER THE SHIFT

Humidity
0 … 100 %

Wind speed
≥ 0

cos, sin
−1.0 … +1.0

All outputs
saturate to 18 bits

`30 °C × 2^8 = 7680 raw.` Floor scaling by arithmetic shift; the C++ reference uses the same rules, so any mismatch is an implementation bug, not forecast error.

SIX PARALLEL LANES

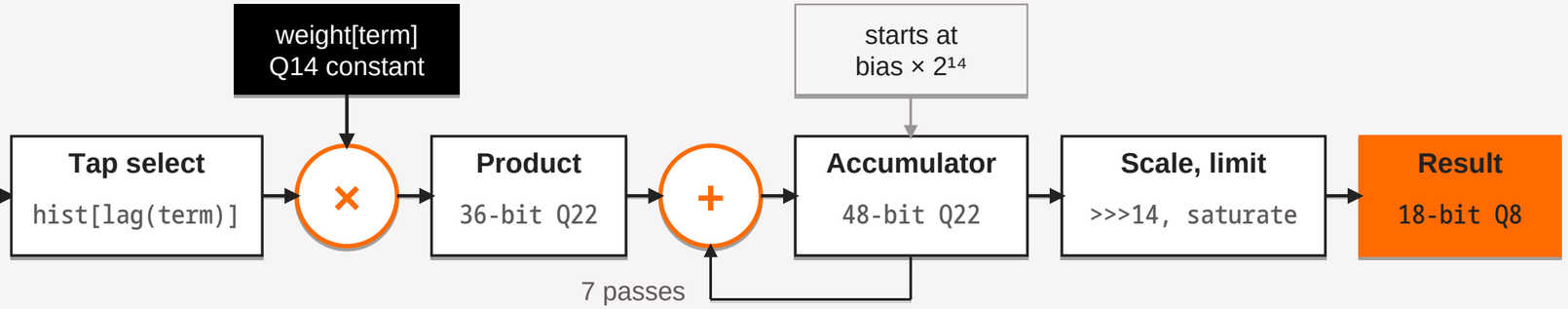
Each lane owns one multiplier and reuses it for all seven taps — 6 multipliers, not 42

Inside one lane `weather_ar_channel.v`

25-frame history, 7 taps (lag in h)

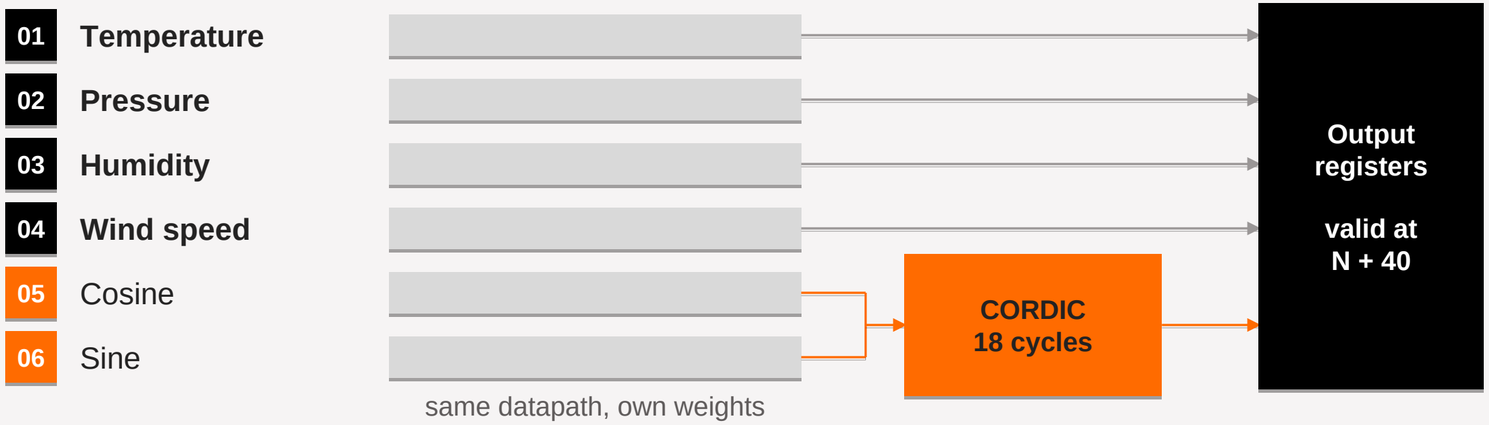


shifts left on each accepted frame



Per tap: operand → product → accumulate, 3 cycles. Seven taps plus accept and publish: lane result at N + 22.

Six copies in parallel `weather_predictor_speed.v`



same datapath, own weights

Handshake

History shifts only when valid and ready are both high; idle clocks change nothing.

Warm-up

Frames 1–24 fill the history. Frame 25 gives the first forecast.

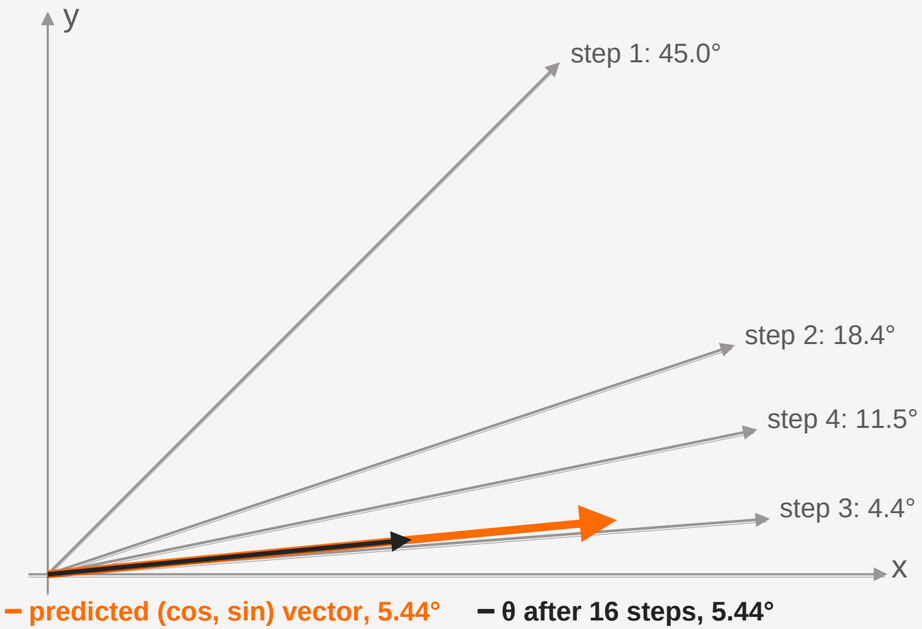
Multipliers

Each 18 × 18 multiplier uses two 9-bit elements: 12 in the Quartus fit.

WIND DIRECTION: CORDIC

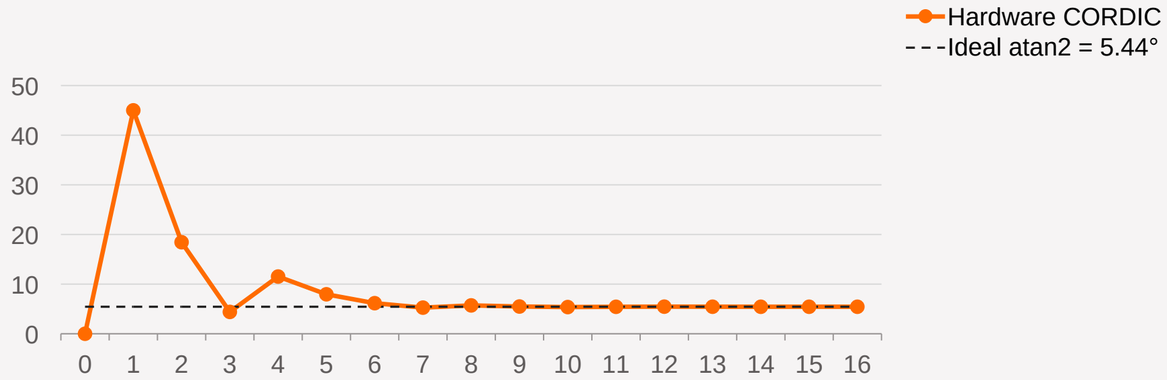
Sixteen shift-and-add rotations turn the predicted (cos, sin) into an angle

Board example: cos θ = 0.902, sin θ = 0.086



Grey arrows: the hardware's angle estimate θ after steps 1–4. It overshoots to 45°, then swings back in halving steps until it lands on the predicted vector.

Accumulated angle θ after each step, degrees



One step i = 0 … 15 in wind_direction_cordic.v

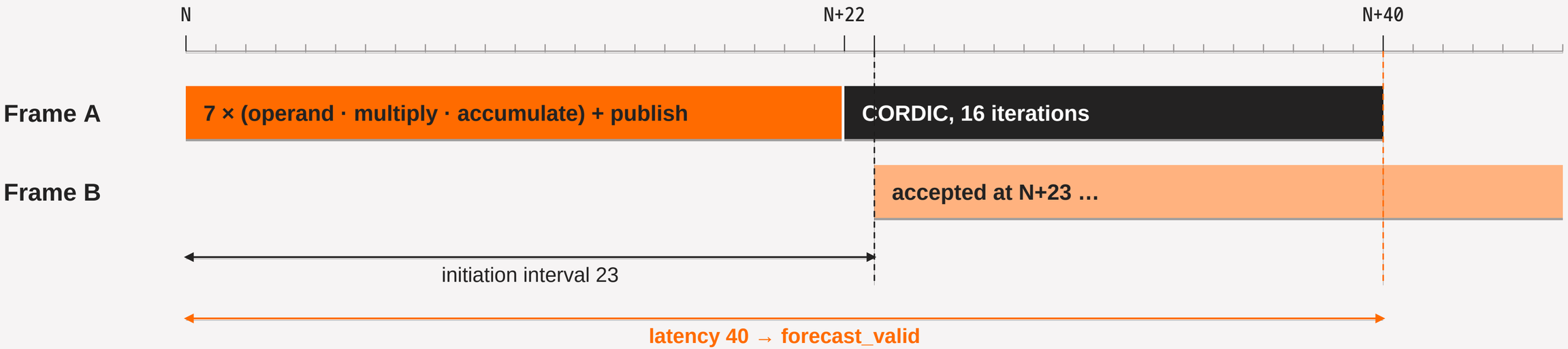
```
y > 0:  x += y>>>i   y -= x>>>i   θ += atan(2^-i)
y < 0:  x -= y>>>i   y += x>>>i   θ -= atan(2^-i)
```

Step 1: y = 0.086 > 0, the true angle is above the estimate 0°, so θ += 45°. Internally (x, y) is turned back by the same 45°, so the next step can test the new estimate. Only shifts, adds and a 16-entry table, no multiplier.

Result: the turns add up to θ = 5.44°, a wind from just east of north. **The board reported 5.4375°, 0.003° from the exact value.** Over every possible input pair the worst error is 0.0056°.

LATENCY VS. INITIATION INTERVAL

A forecast takes 40 cycles; a new frame can enter every 23 cycles



Forecast horizon	Sampling interval	Clock period	Prediction latency	Initiation interval
3 h	**1 h**	**20 ns**	**40 cycles = 0.80 µs**	**23 cycles = 0.46 µs**
data timestamps	NASA POWER hourly	50 MHz board clock	22 lane + 18 CORDIC	lane ready again

VERIFICATION

One C++ golden model, three increasingly physical checks

01

RTL + DPI-C

Verilator, C++ reference called per cycle

1,200 accepted frames · 4 pending forecasts cancelled by reset · warm-up repeated · latency 40

1,124 forecasts checked bit-exact

02

Predictor-core SDF

ModelSim, post-fit netlist + SDF delays, 20 ns clock

288 input frames · predictor core only, not the JTAG transport

264 forecasts checked

03

Physical JTAG

DE2 board at 50 MHz through USB-Blaster

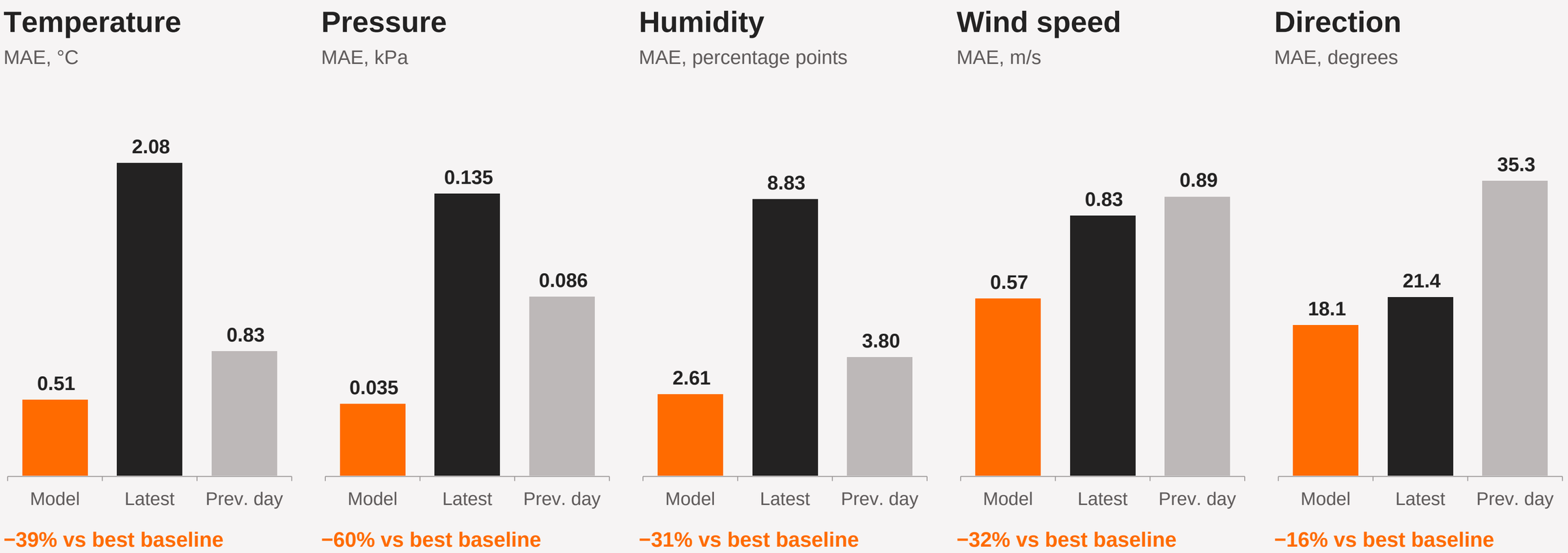
128 recorded frames · scalars and components bit-exact · angle within 5 Q8 units

104 forecasts checked

Agreement with the reference shows the hardware computes the model correctly. Forecast accuracy against real observations is a separate question — next slide.

FORECAST ACCURACY

Lower test error than both baselines for all five quantities (2025)

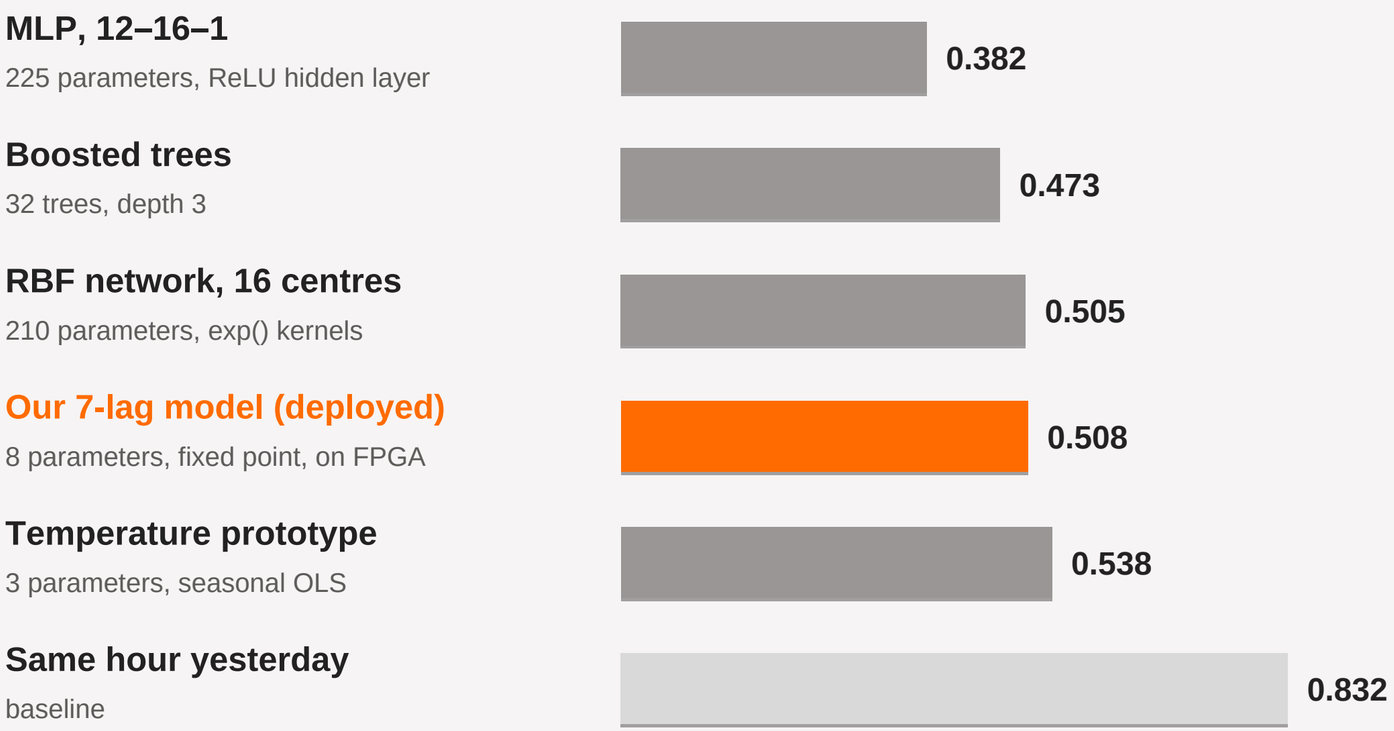


8,757 scalar targets. Direction: 8,593 targets with wind ≥ 0.5 m/s, ideal angle decoding of quantised components.

SIMPLER ON PURPOSE

On the same data, larger models forecast temperature better — we chose the one that fits hardware

Temperature test MAE, 2025, °C — lower is better



WHY WE KEPT IT LINEAR

Accuracy cost

The MLP is 0.13 °C better. The RBF is level with ours (0.003 °C). Ours is still 39% better than the best baseline.

Hardware cost

A 7-lag sum is one multiplier reused 7 times per lane. The MLP needs 16 hidden neurons; the RBF needs exponentials.

Verifiability

Fixed-point sums can be checked bit-for-bit against C++ at every stage.

Same NASA POWER data, same chronological split, 8,757 test targets. Benchmarks are float software models of temperature only; ours is the deployed fixed-point lane.

QUARTUS FIT

Uninstrumented JTAG build, Quartus II 13.0 SP1

Flow Summary	
Flow Status	Successful - Tue Oct 27 20:50:14 2026
Quartus II 32-bit Version	13.0.1 Build 232 06/12/2013 SP 1 SJ Web Edition
Revision Name	weather_jtag
Top-level Entity Name	de2_weather_jtag
Family	Cyclone II
Device	EP2C35F672C6
Timing Models	Final
Total logic elements	6,253 / 33,216 (19 %)
Total combinational functions	4,147 / 33,216 (12 %)
Dedicated logic registers	4,571 / 33,216 (14 %)
Total registers	4571
Total pins	2 / 475 (< 1 %)
Total virtual pins	0
Total memory bits	0 / 483,840 (0 %)
Embedded Multiplier 9-bit elements	12 / 70 (17 %)
Total PLLs	0 / 4 (0 %)

Native Flow Summary pane, cropped; no values changed. Full timing panels: backup B5.

1. **6,253 logic elements**
   19% of 33,216 in the EP2C35F672C6
2. **4,571 registers**
   mostly the six 25-frame history buffers
3. **12 nine-bit multipliers**
   six 18 × 18 multipliers, one per lane

Timing at 50 MHz

setup slack 7.459 ns, hold slack 0.215 ns

79.74 MHz Fmax

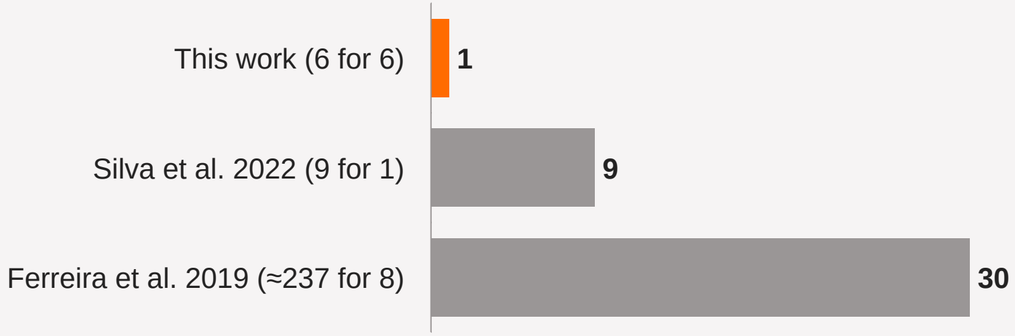
static slow-model estimate; the board was tested at 50 MHz only

HOW IT COMPARES

Fewest multipliers per forecast value, at a similar rate per value

Hardware multipliers per predicted value

18 × 18-class multiplier blocks ÷ values produced in parallel



Time per predicted value, ns

ours: 23-cycle interval ÷ 6 values at 50 MHz



	Logic	Format	Evidence
This work	6,253 LEs	18-bit fixed	board, 50 MHz
Ferreira 2019	≈9,200 LEs	32-bit fixed	“1 cycle”, no Fmax
Silva 2022	3,440 LUT6	32-bit float	synthesis
Vásquez 2013	n/a	fixed point	abstract only

WHAT THIS SHOWS

Cheapest in multipliers: one per value, because each is reused for all 7 taps.

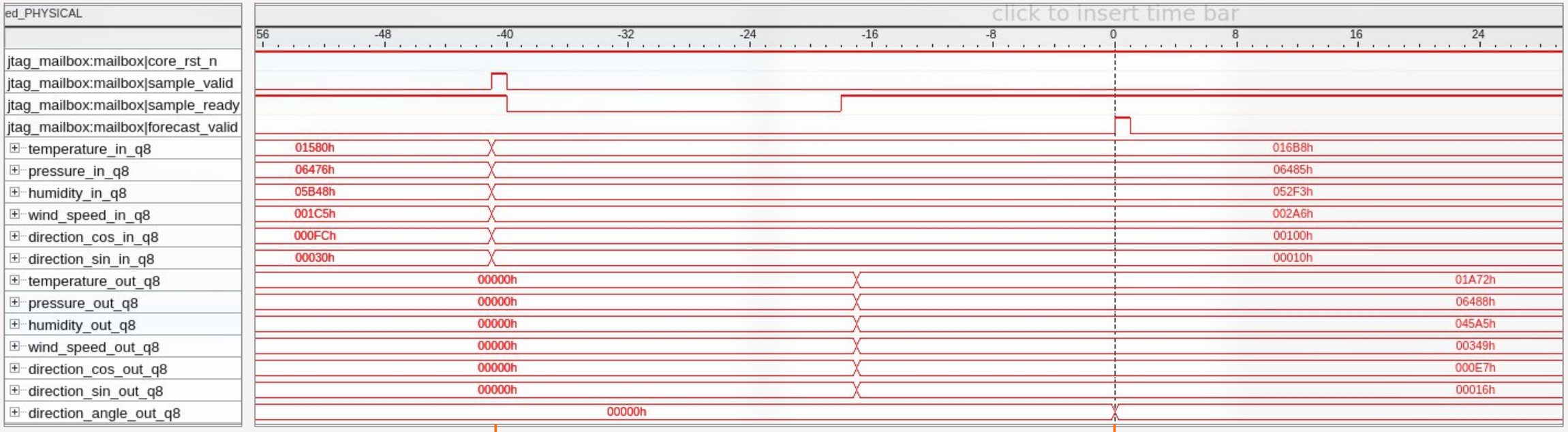
Comparable rate: 77 ns per value vs 40–74 ns published.

The price: 0.80 µs latency per frame, far below the 1-hour budget.

Caveats: different devices and tasks — Ferreira and Silva also fit the regression on chip; Ferreira's figures are rounded percentages (8% LEs, 89% of 532 9-bit multipliers). Full table and citations: backups B8–B9.

ON THE BOARD

SignalTap: one accepted frame, and its forecast 41 samples later



Native capture, cropped

frame accepted · sample 215

forecast_valid · sample 256

Signal	Hex	Raw	Value
`temperature_out_q8`	`01A72h`	`6770`	26.4453125 °C
`direction_angle_out_q8`	`00570h`	`1392`	5.4375°

Replay: 128 frames → 104 forecasts, all matching.

Timing: 41 acquisition intervals = the RTL's 40-cycle post-edge latency.

Separately instrumented build · 512 samples at 20 ns · full view: B6

SCOPE AND LIMITATIONS

What our evidence supports, and what it does not

WHAT WE SHOW

- ✓ Five quantities forecast 3 h ahead from recorded hourly frames
- ✓ Bit-exact agreement with the C++ reference in RTL, the core netlist and on the board
- ✓ 50 MHz operation, 40-cycle latency, 23-cycle initiation interval
- ✓ Lower 2025 test MAE than persistence and same-hour-yesterday

WHAT WE DO NOT CLAIM

- ✕ Training on the FPGA, live sensors, or other sites and years
- ✕ Rainfall probability, or LED / LCD output
- ✕ A measured Fmax or power: 79.74 MHz is a static estimate
- ✕ Timing simulation of the full JTAG transport (SDF covers the core)

NEXT STEPS

01 Feed live observations instead of replayed frames

02 Retrain and test on other sites and years

03 Measure power and sweep the clock on the board

04 Add a rainfall-probability output

THANK YOU

- Seven-lag fixed-point regression forecasts five weather quantities three hours ahead.
- Six parallel lanes and a CORDIC give a 0.80 µs latency and a 23-cycle interval at 50 MHz.
- The same C++ reference agrees with RTL, the post-fit core and 104 forecasts from the DE2 board.

QUESTIONS?

BACKUP B1

ORIGINAL TEMPERATURE MODEL

$$\hat{T}(t+3) = T(t-21) + a\cdot[T(t) - T(t-24)] + b\cdot[T(t) - T(t-3)] + c$$

Parameter	Fitted value
a	0.7454870439
b	0.0091494450
c	0.0011295817 °C

T(t−21) is the same hour yesterday relative to the target (t+3). a corrects with the daily change, b with the recent three-hour change, c is a constant.

One lane, signed 16-bit Q8.

Test MAE (2025, fixed point): 0.538017 °C prototype → 0.507906 °C current weather temperature lane.

BACKUP B2

COEFFICIENTS AND A WORKED CALCULATION

Lag (h)	Input Q8	Weight Q14	Product
0	7680	20035	153868800
1	7552	−9335	−70497920
3	7168	1621	11619328
6	6912	−205	−1416960
12	6656	15	99840
21	7936	13908	110373888
24	7424	−10382	−77075968

Bias: 310 (Q8) × 16384

Sum S: 132050048

⌊S / 16384⌋ = 8059

Value: 8059 / 256 = 31.48047 °C

Illustrative input set (temperature lane); deployed weights_q14 and bias_q8.

BACKUP B3

FULL TEST METRICS, 2025

Target	Model MAE	Model RMSE	Latest-value MAE	Same-hour-yesterday MAE
Temperature (°C)	0.507906	0.730939	2.080369	0.832280
Pressure (kPa)	0.034600	0.044638	0.135167	0.085775
Humidity (percentage points)	2.614292	3.685767	8.831186	3.804237
Wind speed (m/s)	0.566878	0.750218	0.830494	0.891217
Direction (degrees)	18.084598	30.201056	21.423810	35.292273

Model = selected_fixed (deployed fixed point). 8,757 scalar targets; direction: 8,593 targets with true wind speed ≥ 0.5 m/s, ideal atan2 of quantised components. CORDIC maximum conversion error 0.005635° (exhaustive).

BACKUP B4

CYCLE SCHEDULE OF ONE LANE

Edge	Lane state	Operation
N	0 → 1	Accept frame (valid ∧ ready), shift history, load aligned bias
N + 1	1	Capture operand hist[lag 0]
N + 2	2	Capture 36-bit product
N + 3	3	Add product to 48-bit accumulator
N + 4 … N + 21	1 → 2 → 3	Repeat for the remaining six taps
N + 22	4	Shift >>> 14, limit, saturate, publish; lanes ready
N + 23	0	Earliest next accepted frame
N + 22 … N + 40	CORDIC	Angle unit runs; system forecast_valid at N + 40

Synchronous active-low reset clears history eligibility and cancels pending work; a full new warm-up is required.

BACKUP B5

NATIVE TIMING REPORTS

Multicorner Timing Analysis Summary

	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	Worst-case Slack	7.459	0.215	N/A	N/A	7.981
1	CLOCK_50	7.459	0.215	N/A	N/A	7.981
2	altera_reserved_tck	N/A	N/A	N/A	N/A	97.778
2	Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	CLOCK_50	0.000	0.000	N/A	N/A	0.000
2	altera_reserved_tck	N/A	N/A	N/A	N/A	0.000

Slow Model Fmax Summary

	Fmax	Restricted Fmax	Clock Name	Note
1	79.74 MHz	79.74 MHz	CLOCK_50	

Physical test clock: 50 MHz

Setup slack: 7.459 ns

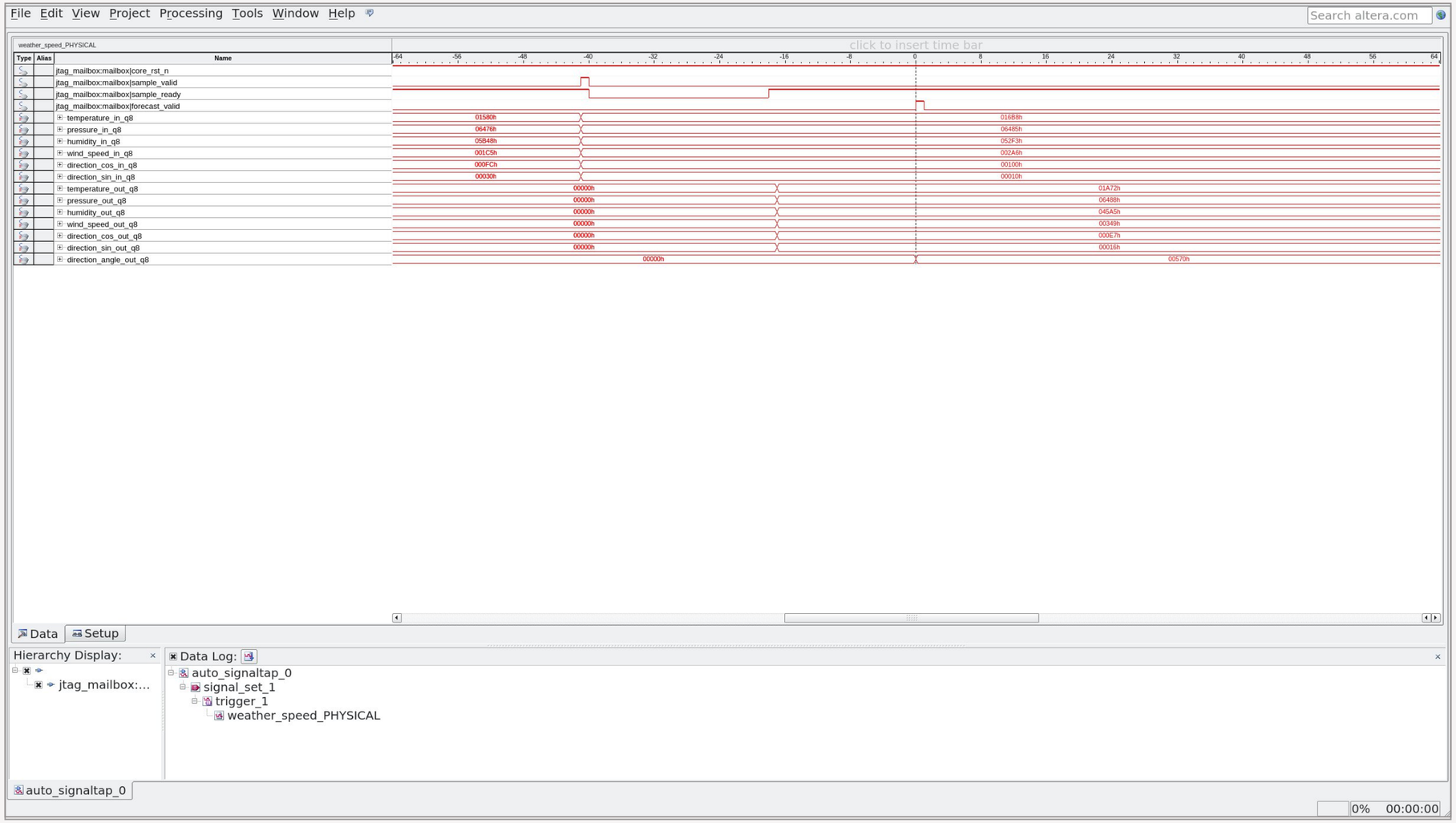
Hold slack: 0.215 ns

Slow-model Fmax: 79.74 MHz (estimate)

TimeQuest panels of the uninstrumented weather-speed JTAG build, cropped; no values changed.

BACKUP B6

ORIGINAL SIGNALTAP ACQUISITION



BACKUP B7

JTAG MAILBOX PROTOCOL

01 Stage payload with the old request bit

02 Invert the request bit

03 Wait for the acknowledgement toggle

04 Read a stable probe word

One outstanding request at a time. Warm-up requests are acknowledged without a forecast. The latest forecast stays latched until the next forecast or reset.

Bits	Host → FPGA (128-bit source)
0	Request toggle, one change per frame
1	Reset / clear (level)
109:2	Six signed 18-bit Q8 channels
127:110	Reserved, zero

Bits	FPGA → host (256-bit probe)
107:0	Six Q8 forecasts
125:108	Direction angle, degrees Q8
143:128 · 159:144	Accepted / forecast counters
160 · 162 · 163	Ack toggle · result valid · direction defined
255:224	Signature 46504741h

BACKUP B8

PUBLISHED FPGA PREDICTORS

Design	Task	Device · arithmetic	Resources	Timing
This work	5 quantities, 6 lanes × 7 taps	Cyclone II EP2C35 · 18-bit fixed	6,253 LEs · 12 × 9-bit mult.	0.80 µs RTL latency at 50 MHz board clock
Vásquez et al. 2013	Temperature, MLP network	Xilinx · fixed point	not in abstract	74 ns max. delay (abstract)
Ferreira et al. 2019	Linear-regression fit + 8 predictions	Cyclone IV EP4CE115 · 32-bit fixed	8% of 114,480 LEs · 89% of 532 mult.	“one clock cycle”, no Fmax
Silva et al. 2022	Rolling linear-regression predictor	Virtex-6 · 32-bit float	3,440 LUTs · 9 mult.	40.25 ns processing (synthesis)
Royer et al. 2011	Block linear-regression fit	Xilinx DSP48E · fixed point	66 slices · 5 DSP48E	458 MHz Fmax (estimate)

LEs, LUTs and slices are different units, and the timing evidence ranges from board tests to estimates: context, not a ranking. Citations: backup B9.

BACKUP B9

SOURCES AND CREDITS

Model and evaluation	versions/weather_v2/models/model.json · versions/weather_v2/reports/metrics.csv · versions/weather_v2/data/weather.csv · software/models_cpp/model.json · software/benchmarks/
Hardware	versions/weather_v2/rtl/*.v · versions/weather_v2/tb/tb_weather.sv · weather_reference.cpp · hardware/jtag/
Evidence	docs/reports/JTAG_IMPLEMENTATION_REPORT.md · docs/reports/weather_speed_modelsim_timing.json · docs/reports/defense_session/ (screenshots, replay log, SignalTap)
Related work	Vásquez et al. 2013, doi:10.1007/s12559-012-9158-z · Ferreira et al. 2019, doi:10.1145/3338852.3339853 · Silva et al. 2022, doi:10.3390/s22093556 · Royer et al., DCIS 2011
Template	“Automotive Training Center” by SlidesCarnival (slidescarnival.com), CC BY 4.0; adapted, stock photos removed